Adversarial Möbius Completions and Signed Displacement Cuts: A Combinatorial Framework with Exact Finite Results

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Abstract

We study a finite, deterministic completion problem for the Möbius function. At scale N, the non-squarefree support is fixed, selected prime signs may be fixed, and the remaining prime signs are treated adversarially. The endpoint envelope

$$B_N(\mathcal{I}) = \max_{x \in \mathcal{C}_N(\mathcal{I})} \left| \sum_{n \le N} x_n \right|$$

separates the amount of information supplied from the worst endpoint still compatible with that information. The first arithmetic layers admit exact closed forms: unrestricted squarefree signs and free multiplicative prime signs both have envelope $Q(N)$, while fixing the sign at 2 gives

$$B_N = Q_{\mathrm{odd}}(N) - Q_{\mathrm{odd}}(N/2) \sim \frac{2}{\pi^2} N.$$

Thus a very large reduction in the number of completions need not change the endpoint exponent.

The second part of the paper develops a finite combinatorial representation of short-correlation comparisons. For

$$C_h(x) = \sum_{n \le N-h} x_n x_{n+h}, \qquad d_h(x) = C_h(x) - C_h(\mu),$$

the discrepancy set $D(x) = \{n : x_n \neq \mu(n)\}$ defines a cut in a signed displacement graph G_h. Exactly,

$$\frac{d_h(x)}{2} = \mathrm{Cut}_{G_h}(D(x)).$$

Because admissible discrepancy sets arise from prime-sign assignments, they form the image of a binary linear code. Each displacement edge yields a parity form $q \cdot b$, and its sign is determined by the parity identity

$$-\mu(n)\mu(n+h) = (-1)^{|q|+h+1}.$$

Hence every first-witness condition is an explicit threshold of low-order parity tests on the free prime-sign cube. An exhaustive computation at $N = 110$ over 2^{28} completions gives 9,311,936 strict square-root excursions, a survivor chain ending in $13 \to 2 \to 0$ at lags $9, 10, 11$, and two hardest states with first witness $h_* = 11$. Complete signed-cut data for all 109 lags are recorded.

1 Scope and notation

Let

$$M(N) = \sum_{n \le N} \mu(n)$$

and let

$$Q(N) = \#\{n \le N : \mu(n) \neq 0\}.$$

The finite program begins with a family of words

$$x = (x_1, \dots, x_N) \in \{-1, 0, 1\}^N$$

and progressively imposes exact arithmetic information. If $\mathcal{I}$ denotes the declared information and closure rules, write

$$\mathcal{C}_N(\mathcal{I}) = \{x : \ x \text{ is compatible with } \mathcal{I}\}$$

and

$$\boxed{B_N(\mathcal{I}) = \max_{x \in \mathcal{C}_N(\mathcal{I})} \left| \sum_{n \le N} x_n \right|.}$$

The object is deterministic: a single surviving completion matters, irrespective of its proportion inside the state space.

The strict square-root excursion class is

$$\boxed{\mathrm{RHC}_N = \{x : |S_x(N)| > \sqrt{N}\}}, \qquad S_x(N) = \sum_{n \le N} x_n.$$

For $\delta \ge 0$ define

$$\boxed{\mathrm{RHF}_{N,\delta} = \{x : |S_x(N)| > N^{1/2+\delta}\}, \qquad \mathrm{RHF}_{N,0} = \mathrm{RHC}_N.}$$

The distinction is finite and exact. In particular, the false Mertens inequality $|M(N)| \le \sqrt{N}$ cannot hold for every N, whereas the Riemann hypothesis is equivalent to

$$M(N) = O_\varepsilon(N^{1/2+\varepsilon}) \qquad (\forall\, \varepsilon > 0).$$

2 Literature boundary

The neighboring literature is substantial, but it addresses different objects.

Vandehey introduced P-multiplicative functions, imposing $f(pn) = f(p)f(n)$ only for primes in a prescribed finite set P, and studied bounded partial sums in that weakened multiplicative setting [1]. Thus finite prime prescriptions and partial multiplicativity are established objects.

Aymone and Sidoravicius study random multiplicative functions supported on the squarefree integers whose prime values are independent $\{\pm 1\}$ variables biased toward the Möbius signs, including an RH-equivalent probabilistic statement for a family of such models [2]. Koukoulopoulos and Soundararajan prove inverse structural results for multiplicative functions with very small partial sums [3]. Matomäki, Radziwiłł, and Tao establish local sign-pattern results for Liouville and Möbius [4]. These works delimit the classical multiplicative-function side of the problem.

Aperiodic autocorrelation of finite binary sequences is also classical; Willms, for example, relates the full autocorrelation vector to run structure [5]. Signed graphs and signed cuts form a mature graph-theoretic framework [6, 7]. Consequently, neither "autocorrelation" nor "signed cut" by itself constitutes a novelty claim.

The specific finite object isolated below is more restrictive: a simultaneous family of signed displacement graphs whose edge signs are inherited from the finite Möbius word, with cuts restricted to discrepancy sets generated by squarefree multiplicative prime-sign completions. Targeted searches using combinations of Möbius autocorrelation, signed cuts, displacement graphs, multiplicative completions,

and first-lag witnesses did not locate this exact formulation. This negative search is a bibliographic boundary, not a theorem of novelty.

For computational context, Hurst computed all values of $M(x)$ through 10^{16} and later isolated values through 10^{25} [8, 9]. Those computations emphasize the distinction between computing the true Möbius word and exhaustively solving a completion problem over exponentially many compatible words.

3 Exact endpoint envelopes for the first arithmetic layers

3.1 Squarefree support and free multiplicativity

Proposition 3.1. *Let the nonsquarefree coordinates be fixed to zero and every squarefree coordinate be free in* $\{\pm 1\}$*. Then*

$$\boxed{B_N^{\mathrm{sqf}} = Q(N)}.$$

If instead the squarefree word is required to be multiplicative and all prime signs are free, then

$$\boxed{B_N^{\mathrm{mult}} = Q(N)}.$$

Proof. The squarefree-support maximum is attained by assigning $+1$ to every squarefree coordinate. Under multiplicativity the same word is obtained by setting every prime sign to $+1$; every squarefree value is then $+1$. □

Since

$$Q(N) \sim \frac{6}{\pi^2} N,$$

both envelopes have exponent one. Multiplicativity can therefore collapse the state count from exponential in $Q(N)$ to exponential in $\pi(N)$ without changing the extremal envelope.

3.2 Fixing the sign at 2

Let f be squarefree-supported and multiplicative, with

$$f(2) = -1, \qquad f(p) = \sigma_p \in \{\pm 1\} \quad (p \geq 3).$$

Theorem 3.2. *For every* N*,*

$$\boxed{B_N^{\mu(2)} = Q_{\mathrm{odd}}(N) - Q_{\mathrm{odd}}(N/2) = Q_{\mathrm{odd}}(N) - Q_{\mathrm{even}}(N).}$$

Consequently

$$\boxed{B_N^{\mu(2)} \sim \frac{2}{\pi^2} N.}$$

Proof. Every odd squarefree $m \leq N/2$ is paired with $2m$, and

$$f(2m) = f(2)f(m) = -f(m).$$

These pairs cancel. Hence

$$S_f(N) = \sum_{\substack{N/2 < m \leq N \\ m \text{ odd, squarefree}}} f(m).$$

All remaining terms are simultaneously made $+1$ by choosing the odd-prime signs appropriately, so the maximum is their count. Finally, the even squarefree integers up to N are in bijection with odd squarefree $m \le N/2$ via $m \mapsto 2m$. The standard densities are

$$Q_{\text{odd}}(N) \sim \frac{4}{\pi^2} N, \qquad Q_{\text{even}}(N) \sim \frac{2}{\pi^2} N.$$

□

Thus the finite drop

$$16 \longrightarrow 6 \qquad (N = 25)$$

is the finite manifestation of the constant change

$$\frac{6}{\pi^2} \longrightarrow \frac{2}{\pi^2},$$

an asymptotic factor of three, not an exponent contraction.

3.3 Finite exponent bias

If

$$B_N = cN^\theta,$$

then

$$\boxed{E_N := \frac{\log B_N}{\log N} - \frac{1}{2} = \theta - \frac{1}{2} + \frac{\log c}{\log N}.}$$

Thus E_N contains a large finite-size bias when $c \neq 1$. The appropriate scaling diagnostics are

$$R_N = \frac{B_N}{N}$$

and a local log–log slope

$$\boxed{\theta_N^{(a)} = \frac{\log B_{aN} - \log B_N}{\log a}.}$$

For the fixed-$\mu(2)$ layer:

N	$B_N^{\mu(2)}$	$B_N^{\mu(2)}/N$	decade slope
10^2	21	0.210000	0.978811
10^3	200	0.200000	1.006252
10^4	2029	0.202900	0.999357
10^5	20260	0.202600	1.000099
10^6	202646	0.202646	–

4 Two distinct prime-information models

The two following models are not interchangeable.

4.1 Model A: prescribed prime signs without residual propagation

Revealed prime signs are fixed and unrevealed prime signs remain independent. No primality statement is inferred from a residual factor.

At $N = 25$:

revealed signs	states	B_{25}	RHC	$\mathrm{RHF}_{25,0.10}$
$\{2\}$	256	6	8	0
$\{2,3\}$	128	6	4	0
$\{2,3,5\}$	64	6	2	0

Proposition 4.1 (Prime-singleton obstruction)**.** *Suppose k unrevealed primes lie in $(N/2, N]$. Then*

$$B_N \geq k.$$

If the true Möbius completion is present in the admissible set, then

$$\boxed{B_N \geq 2k - |M(N)|}.$$

Proof. Each such prime occurs only at its own coordinate. After the other variables are fixed,

$$S_N = C + \sum_{p \in P_{\mathrm{free}}} \sigma_p,$$

so the maximum over these signs is $|C| + k \geq k$. Starting from the true word and flipping all k prime signs changes the endpoint by $2k$, whence the second inequality. □

If all upper-half primes are free,

$$k = \pi(N) - \pi(N/2) \sim \frac{N}{2 \log N},$$

so this model retains an $N^{1-o(1)}$ endpoint obstruction.

4.2 Model B: residual closure by P_r

Let

$$P_r = \{p_1, \ldots, p_r\}.$$

Strip known prime factors from n. Repeated known factors force zero; residual 1 fixes the sign; a residual $q \leq p_r^2$ with no factor in P_r is forced prime; larger unresolved residual cores are shared ternary variables.

At $N = 25$:

P_r	states	B_{25}	RHC	$\mathrm{RHF}_{25,0.10}$
$\{2\}$	$3^{11} = 177147$	7	1296	162
$\{2,3\}$	$3^6 = 729$	7	18	3
$\{2,3,5\}$	1	2	0	0

The endpoint has the exact form

$$S_N = A_r(N) + \sum_c w_{r,N}(c) z_c, \qquad z_c \in \{-1, 0, 1\},$$

with

$$\boxed{w_{r,N}(c) = \sum_{\substack{d \mid P_r^{\#} \\ dc \le N}} \mu(d).}$$

Hence

$$\boxed{B_r^E(N) = |A_r(N)| + \sum_c |w_{r,N}(c)|}$$

and the endpoint distribution is encoded by

$$x^{A_r(N)} \prod_c (x^{-w_c} + 1 + x^{w_c}).$$

Proposition 4.2 (Residual singleton obstruction)**.** *Every unresolved core*

$$c > \max(N/2, p_r^2), \qquad (c, P_r^{\#}) = 1$$

appears in only one endpoint coordinate. Therefore

$$\boxed{B_{N,P_r} \ge L_r(N) := \#\{\, \max(N/2, p_r^2) < c \le N : (c, P_r^{\#}) = 1 \,\}.}$$

5 Three exact combinatorial barriers

5.1 A terminal global identity

Let f be squarefree-supported and multiplicative, with $f(p) \in \{\pm 1\}$. If

$$\sum_{d \le N} f(d) \left\lfloor \frac{N}{d} \right\rfloor = 1,$$

then, writing $g = \mathbf{1} * f$,

$$g(n) = \prod_{p \mid n} (1 + f(p)) \ge 0$$

and

$$\sum_{n \le N} g(n) = 1.$$

Since $g(1) = 1$, one obtains $g(n) = 0$ for every $2 \le n \le N$, hence

$$f(p) = -1 \qquad (p \le N).$$

Thus this identity is terminal in the prime-sign model: it determines the entire prime-sign prefix.

5.2 Support-transport lemma

Lemma 5.1. *Let U be a target coordinate set and let A_j be constraint rows. Suppose*

$$d_U = \max_j |\operatorname{supp}(A_j) \cap U|.$$

If a dual combination y uses at most s rows, then

$$\boxed{\|\mathbf{1}_U - (A^T y)_U\|_1 \ge |U| - s d_U.}$$

Proof. The union of the selected row supports covers at most sd_U coordinates of U. At every uncovered coordinate, $(A^T y)_u = 0$, so the residual against $\mathbf{1}_U$ contributes exactly one. □

For divisor-local rows on the upper-half squarefree set, $d_U = 1$. For pointwise translational rows of bounded width h, $d_U = O(h)$. This is a coverage obstruction; it does not apply to genuinely global rows.

5.3 Aggregated divisor identities and inversion

From

$$\sum_{d|n} \mu(d) = \delta_{n1},$$

multiply by arbitrary coefficients a_n and aggregate:

$$\sum_{d \le N} \mu(d) \sum_{m \le N/d} a_{dm} = a_1.$$

Put

$$c_d = \sum_{m \le N/d} a_{dm}.$$

Then

$$\boxed{M(N) = a_1 + \sum_{d \le N} \mu(d)(1 - c_d)}$$

and

$$|M(N)| \le |a_1| + \sum_{\substack{d \le N \\ d \text{ squarefree}}} |1 - c_d|.$$

The transform is exactly invertible:

$$\boxed{a_n = \sum_{k \le N/n} \mu(k) c_{nk}.}$$

In particular, the exact target $c_d \equiv 1$ gives

$$a_n = M(\lfloor N/n \rfloor), \qquad a_1 = M(N).$$

If $a_n = 0$ for $n > K$, then $c_d = 0$ for $d > K$, and therefore the squarefree ℓ^1 defect is at least

$$Q(N) - Q(K).$$

6 The fixed-σ_2 multiplicative cube

From now on,

$$x_1 = 1, \qquad x_n = 0 \quad (n \text{ nonsquarefree}),$$

and, for squarefree n,

$$\boxed{x_n = \prod_{p|n} \sigma_p, \qquad \sigma_2 = -1, \quad \sigma_p \in \{\pm 1\} \ (p \ge 3).}$$

Thus

$$|\Omega_N| = 2^{\pi(N)-1}.$$

The true Möbius word is the all-minus prime assignment.

At $N = 25$,

$$\boxed{S_{25} = \sigma_{13} + \sigma_{17} + \sigma_{19} + \sigma_{23} + \sigma_3\sigma_5 + \sigma_3\sigma_7.}$$

The variable σ_{11} is silent. The eight RHC extremizers are described by

$$\sigma_{13} = \sigma_{17} = \sigma_{19} = \sigma_{23} = \tau, \qquad \sigma_5 = \sigma_7 = \tau\sigma_3,$$

with $\tau, \sigma_3, \sigma_{11} \in \{\pm 1\}$.

The first multiplicative dependence among the active endpoint monomials appears at $N = 65$:

$$\boxed{f(33)f(39)f(55)f(65) = 1.}$$

6.1 Exact center of the finite sign cube

Equip the free odd-prime signs with the uniform product measure.

Proposition 6.1. *For $N \geq 2$,*

$$\boxed{\mathbb{E}S_N = 0.}$$

Moreover,

$$\boxed{\mathbb{E}C_h = \begin{cases} -1, & h \text{ odd and squarefree, } 2h \leq N, \\ 0, & \text{otherwise.} \end{cases}}$$

Proof. The terms $x_1 = 1$ and $x_2 = -1$ cancel in expectation. Every other squarefree n contains at least one free odd prime, so $\mathbb{E}x_n = 0$.

For $\mathbb{E}[x_n x_{n+h}]$ to be nonzero, every free odd prime must occur an even number of times across the two squarefree integers. Hence n and $n + h$ must have the same set of odd prime divisors. Since they are distinct squarefree integers, they can then differ only by a factor of 2. As $n + h > n$, necessarily $n = h$ and $n + h = 2h$, with h odd squarefree. In that case $x_h x_{2h} = -1$. □

7 Short correlations and the first witness

Define

$$C_h(x; N) = \sum_{n \leq N-h} x_n x_{n+h}, \qquad 1 \leq h < N,$$

and

$$\boxed{d_h(x; N) = C_h(x; N) - C_h(\mu; N).}$$

The first positive witness is

$$\boxed{h_*(x; N) = \min\{h \geq 1 : d_h(x; N) > 0\},}$$

when it exists.

All admissible words have the same zero support as μ, so

$$S_x(N)^2 = \sum_{n \leq N} x_n^2 + 2 \sum_{h=1}^{N-1} C_h(x; N).$$

Subtracting the Möbius identity gives

$$\boxed{\sum_{h=1}^{N-1} d_h(x;N) = \frac{S_x(N)^2 - M(N)^2}{2}.}$$

Thus $|S_x(N)| > |M(N)|$ forces at least one positive d_h. The finite combinatorial question is the position of the first such h.

For a state with witness h_* define

$$D^-(x) = -\sum_{h<h_*} d_h(x) \geq 0, \qquad J(x) = d_{h_*}(x) > 0.$$

These are exact descriptive quantities; D^- alone does not determine h_*, as the $N = 110$ data below show.

8 Signed displacement cuts

8.1 Cut identity

Let

$$V_N = \{n \leq N : \mu(n) \neq 0\}.$$

For each h, form the graph

$$G_h(N) = (V_N, E_h, w_h),$$

where

$$E_h = \{\{n, n+h\} : 1 \leq n \leq N-h,\ n, n+h \in V_N\}$$

and

$$w_h(\{n, n+h\}) = -\mu(n)\mu(n+h) \in \{\pm 1\}.$$

For an admissible completion x, define its discrepancy set

$$\boxed{D(x) = \{n \in V_N : x_n \neq \mu(n)\}.}$$

Because the zero support is fixed,

$$x_n = \begin{cases} -\mu(n), & n \in D(x), \\ \mu(n), & n \notin D(x). \end{cases}$$

Let $\partial_h D$ be the set of edges in E_h with exactly one endpoint in D, and define the signed cut

$$\mathrm{Cut}_h(D) = \sum_{e \in \partial_h D} w_h(e).$$

Theorem 8.1 (Signed displacement-cut identity)**.** *For every admissible completion and every $1 \leq h < N$,*

$$\boxed{\frac{C_h(x) - C_h(\mu)}{2} = \mathrm{Cut}_h(D(x)).}$$

Equivalently, if

$$A_h(x) = \#\{e \in \partial_h D : w_h(e) = +1\}, \qquad B_h(x) = \#\{e \in \partial_h D : w_h(e) = -1\},$$

then

$$\boxed{d_h(x) = 2(A_h(x) - B_h(x))}$$

and

$$\boxed{h_*(x) = \min\{h : A_h(x) > B_h(x)\}.}$$

Proof. If both endpoints of an edge are simultaneously flipped, or neither is flipped, the product $x_n x_{n+h}$ equals $\mu(n)\mu(n+h)$. If exactly one endpoint is flipped, the product changes sign, so

$$x_n x_{n+h} - \mu(n)\mu(n+h) = -2\mu(n)\mu(n+h) = 2w_h(e).$$

Summing over crossing edges proves the identity. □

For every h, the underlying graph is a disjoint union of fragments of arithmetic progressions

$$r,\ r+h,\ r+2h, \dots,$$

hence a forest of paths. If $h > N/2$, every component contains at most one edge, so G_h is a matching.

8.2 Multiplicative admissibility as a binary code

Index the free odd primes by p. Write

$$b_p = \begin{cases} 0, & \sigma_p = -1, \\ 1, & \sigma_p = +1. \end{cases}$$

For squarefree n,

$$\boxed{\mathbf{1}_{D(x)}(n) \equiv \sum_{\substack{p \mid n \\ p \text{ odd}}} b_p \pmod 2.}$$

If A is the binary incidence matrix

$$A_{n,p} = \mathbf{1}_{p \mid n} \pmod 2,$$

then

$$\boxed{\mathbf{1}_D = Ab \pmod 2.}$$

Thus admissible cuts are not arbitrary cuts of G_h; they belong to the image of a specific binary linear code generated by prime divisibility.

For an edge $e = \{n, n+h\}$ define

$$\boxed{q_e = A_n + A_{n+h} \in \mathbb{F}_2^{\pi(N)-1}.}$$

Then

$$e \in \partial_h D \iff q_e \cdot b = 1 \pmod 2.$$

Proposition 8.2 (Parity of the edge sign)**.** *For every displacement edge,*

$$\boxed{w_h(e) = (-1)^{|q_e|+h+1},}$$

where $|q_e|$ is the Hamming weight of q_e.

Proof. For squarefree endpoints,

$$\mu(n)\mu(n+h) = (-1)^{\omega(n)+\omega(n+h)}.$$

Every common odd prime contributes twice and disappears modulo 2; the remaining odd-prime parity is $|q_e|$. The prime 2 lies in the symmetric difference of the prime supports exactly when n and $n+h$ have opposite parity, equivalently when h is odd. Hence

$$\omega(n) + \omega(n+h) \equiv |q_e| + h \pmod 2,$$

and multiplying by the leading minus sign gives the formula. □

Group edges with the same parity form:

$$m_h(q) = \#\{e \in E_h : q_e = q\}.$$

Then

$$\boxed{\frac{d_h(b)}{2} = \sum_{\substack{q \\ q\cdot b=1}} m_h(q)(-1)^{|q|+h+1}.}$$

Therefore

$$\boxed{h_*(b) = \min\left\{ h : \sum_{q\cdot b=1} m_h(q)(-1)^{|q|+h+1} > 0 \right\}.}$$

This is an exact finite threshold system of parity tests on the prime-sign cube.

9 Exact correlation benchmark at $N = 25$

At $N = 25$ the fixed-σ_2 family contains 256 states. Its endpoint distribution is

S	-6	-4	-2	0	2	4	6
count	4	24	60	80	60	24	4

so exactly eight states satisfy RHC.

For the true Möbius word,

$$\boxed{(C_1, C_2, C_3) = (-3, -1, -5).}$$

Sequentially imposing these exact correlation values inside the 256-state family gives

$$\boxed{256 \xrightarrow{C_1=-3} 56 \xrightarrow{C_2=-1} 19 \xrightarrow{C_3=-5} 1.}$$

The unique final state is the Möbius word. Moreover, every one of the eight RHC states has

$$C_2 \in \{3, 5, 7, 9\},$$

whereas $C_2(\mu) = -1$.

10 Canonical finite benchmark for $1 \le N \le 25$

This section fixes a single reproducible finite benchmark. All counts are exact. The unrestricted ternary layer is computed by coefficient convolution of $(z^{-1} + 1 + z)^N$; the squarefree-support layer fixes $\mu(1) = 1$, fixes zero on nonsquarefree indices, and leaves the remaining squarefree signs free; the multiplicative layer assigns free signs to the primes $p \le N$ and extends multiplicatively on squarefree integers; the last nonterminal layer additionally fixes only $\mu(2) = -1$. No true values $M(N)$ are used in these four layers.

The strict endpoint classes used here are

$$\mathrm{RHC}_N = \{\omega : |S_\omega(N)| > \sqrt{N}\}, \qquad \mathrm{RHF}_{N,\delta} = \{\omega : |S_\omega(N)| > N^{1/2+\delta}\}.$$

The original reference population 3^N is retained in every table. A terminal control layer obtained by imposing the full prime rule $\mu(p) = -1$ for every prime $p \le N$ contains exactly one word and is therefore excluded from the nontrivial filtration.

10.1 Surviving completion counts

N	3^N	squarefree support	multiplicative	multiplicative $+\mu(2) = -1$
1	3	1	1	1
2	9	2	2	1
3	27	4	4	2
4	81	4	4	2
5	243	8	8	4
6	729	16	8	4
7	2,187	32	16	8
8	6,561	32	16	8
9	19,683	32	16	8
10	59,049	64	16	8
11	177,147	128	32	16
12	531,441	128	32	16
13	1,594,323	256	64	32
14	4,782,969	512	64	32
15	14,348,907	1,024	64	32
16	43,046,721	1,024	64	32
17	129,140,163	2,048	128	64
18	387,420,489	2,048	128	64
19	1,162,261,467	4,096	256	128
20	3,486,784,401	4,096	256	128
21	10,460,353,203	8,192	256	128
22	31,381,059,609	16,384	256	128
23	94,143,178,827	32,768	512	256
24	282,429,536,481	32,768	512	256
25	847,288,609,443	32,768	512	256

10.2 Extremal envelope and archival finite effective exponent

	ternary		squarefree		multiplicative		$+\mu(2) = -1$	
N	B_N	E_N	B_N	E_N	B_N	E_N	B_N	E_N
1	1	–	1	–	1	–	1	–
2	2	+0.50000	2	+0.50000	2	+0.50000	0	–
3	3	+0.50000	3	+0.50000	3	+0.50000	1	-0.50000
4	4	+0.50000	3	+0.29248	3	+0.29248	1	-0.50000
5	5	+0.50000	4	+0.36135	4	+0.36135	2	-0.06932
6	6	+0.50000	5	+0.39824	5	+0.39824	1	-0.50000

N	B_0	E_0	B_{sqf}	E_{sqf}	B_{mult}	E_{mult}	B_2	E_2
7	7	+0.50000	6	+0.42078	6	+0.42078	2	-0.14379
8	8	+0.50000	6	+0.36165	6	+0.36165	2	-0.16667
9	9	+0.50000	6	+0.31546	6	+0.31546	2	-0.18454
10	10	+0.50000	7	+0.34510	7	+0.34510	1	-0.50000
11	11	+0.50000	8	+0.36719	8	+0.36719	2	-0.21094
12	12	+0.50000	8	+0.33683	8	+0.33683	2	-0.22106
13	13	+0.50000	9	+0.35663	9	+0.35663	3	-0.07168
14	14	+0.50000	10	+0.37250	10	+0.37250	2	-0.23735
15	15	+0.50000	11	+0.38547	11	+0.38547	3	-0.09432
16	16	+0.50000	11	+0.36486	11	+0.36486	3	-0.10376
17	17	+0.50000	12	+0.37706	12	+0.37706	4	-0.01070
18	18	+0.50000	12	+0.35972	12	+0.35972	4	-0.02038
19	19	+0.50000	13	+0.37112	13	+0.37112	5	+0.04660
20	20	+0.50000	13	+0.35620	13	+0.35620	5	+0.03724
21	21	+0.50000	14	+0.36682	14	+0.36682	6	+0.08852
22	22	+0.50000	15	+0.37610	15	+0.37610	5	+0.02068
23	23	+0.50000	16	+0.38426	16	+0.38426	6	+0.07144
24	24	+0.50000	16	+0.37242	16	+0.37242	6	+0.06379
25	25	+0.50000	16	+0.36135	16	+0.36135	6	+0.05664

10.3 RHC survivor counts

N	3^N	ternary RHC	squarefree RHC	multiplicative RHC	$+\mu(2)$ RHC
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	548	7	3	0
8	6,561	1,854	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	34,938	37	7	0
12	531,441	115,440	37	7	0
13	1,594,323	376,196	46	10	0
14	4,782,969	1,212,672	176	16	0
15	14,348,907	3,874,980	232	10	0
16	43,046,721	7,253,118	232	10	0
17	129,140,163	23,469,114	299	15	0
18	387,420,489	75,346,590	299	15	0
19	1,162,261,467	240,301,818	1,093	53	8
20	3,486,784,401	762,080,202	1,093	53	8
21	10,460,353,203	2,405,119,464	1,471	33	4
22	31,381,059,609	7,558,657,992	4,944	56	8
23	94,143,178,827	23,667,691,934	6,885	82	8
24	282,429,536,481	73,869,142,280	6,885	82	8
25	847,288,609,443	150,937,165,552	6,885	82	8

10.4 $\text{RHF}_{0.05}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
6	729	156	6	2	0
7	2,187	548	7	3	0
8	6,561	846	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	34,938	37	7	0
12	531,441	115,440	37	7	0
13	1,594,323	200,592	46	10	0
14	4,782,969	672,132	56	8	0
15	14,348,907	2,220,600	232	10	0
16	43,046,721	7,253,118	232	10	0
17	129,140,163	23,469,114	299	15	0
18	387,420,489	75,346,590	299	15	0
19	1,162,261,467	141,568,470	378	23	0
20	3,486,784,401	459,240,570	378	23	0
21	10,460,353,203	1,479,280,662	1,471	33	4
22	31,381,059,609	4,736,045,276	1,941	26	0
23	94,143,178,827	15,082,716,810	6,885	82	8
24	282,429,536,481	47,810,887,112	6,885	82	8
25	847,288,609,443	150,937,165,552	6,885	82	8

10.5 $\mathbf{RHF}_{0.10}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	8	1	1	0
4	81	10	1	1	0
5	243	42	1	1	0
6	729	156	6	2	0
7	2,187	226	7	3	0
8	6,561	846	7	3	0
9	19,683	3,014	7	3	0
10	59,049	10,386	8	2	0
11	177,147	16,854	9	3	0
12	531,441	58,812	9	3	0
13	1,594,323	200,592	46	10	0
14	4,782,969	672,132	56	8	0
15	14,348,907	1,157,538	67	5	0
16	43,046,721	3,922,206	67	5	0
17	129,140,163	13,105,098	299	15	0
18	387,420,489	43,282,686	299	15	0
19	1,162,261,467	141,568,470	378	23	0
20	3,486,784,401	257,959,890	378	23	0
21	10,460,353,203	851,645,150	470	17	0
22	31,381,059,609	2,788,013,542	1,941	26	0
23	94,143,178,827	9,060,478,368	2,517	40	0
24	282,429,536,481	29,257,136,160	2,517	40	0
25	847,288,609,443	93,945,019,352	2,517	40	0

10.6 $\mathbf{RHF}_{0.20}$ survivor counts

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
1	3	0	0	0	0
2	9	2	1	1	0
3	27	2	1	1	0

N	3^N	ternary	squarefree	multiplicative	$+\mu(2)$
4	81	10	1	1	0
5	243	12	1	1	0
6	729	56	1	1	0
7	2,187	226	7	3	0
8	6,561	314	1	1	0
9	19,683	1,250	1	1	0
10	59,049	1,782	1	1	0
11	177,147	7,020	9	3	0
12	531,441	26,340	9	3	0
13	1,594,323	39,860	10	4	0
14	4,782,969	149,610	11	3	0
15	14,348,907	544,068	67	5	0
16	43,046,721	1,929,774	67	5	0
17	129,140,163	3,125,622	79	8	0
18	387,420,489	11,148,402	79	8	0
19	1,162,261,467	39,021,138	92	12	0
20	3,486,784,401	64,683,150	92	12	0
21	10,460,353,203	227,686,670	106	11	0
22	31,381,059,609	789,507,890	576	14	0
23	94,143,178,827	2,702,576,730	697	22	0
24	282,429,536,481	4,644,553,260	697	22	0
25	847,288,609,443	15,993,177,852	697	22	0

10.7 Immediate finite observations

The benchmark separates reduction in state count from reduction in the extremal envelope. The E_N columns are retained only as exact finite diagnostics; the closed forms above show that all four baseline envelopes have exponent one. At $N = 25$ the squarefree-support layer has 32768 words and the multiplicative layer has only 512, yet both have $B_{25} = 16$. Fixing $\sigma_2 = -1$ changes the constant sharply, from 16 to 6, but the closed form above shows that this layer still has linear asymptotic growth. At $N = 25$ it leaves eight RHC states and no $\mathrm{RHF}_{25,0.10}$ state. The short-correlation benchmark adds a different kind of information: the first three correlations identify the true word uniquely among the 256 fixed-σ_2 states, and lag 2 separates it from all eight RHC states.

11 Validated exhaustive computation at $N = 110$

The fixed-σ_2 family has 28 free odd-prime signs, hence

$$|\Omega_{110}| = 2^{28} = 268{,}435{,}456.$$

The exhaustive engine was cross-checked against an independent brute-force implementation for every state for $N \le 25$, reproducing the $N = 25$ endpoint distribution and RHC count exactly. The $N = 110$ run used deterministic sampled audits in addition to the independent reconstruction of retained states.

For RHC,

$$\boxed{\#\mathrm{RHC}_{110} = 9{,}311{,}936}$$

and

$$\boxed{H_{\max,0}(110) = 11.}$$

The survivor cascade is:

H	witnesses at H	survivors after H	survivor fraction
0	0	9,311,936	1
1	1,809,348	7,502,588	0.8056958295
2	6,814,779	687,809	0.07386315799
3	672,761	15,048	0.001615990488
4	11,781	3,267	0.0003508400401
5	1,801	1,466	0.0001574323535
6	926	540	5.799008928e-05
7	315	225	2.41625372e-05
8	157	68	7.302455687e-06
9	55	13	1.396057705e-06
10	11	2	2.147781084e-07
11	2	0	0

The same enumeration supplies the following RHF thresholds:

δ	threshold	RHF states	$H_{\max}$	H_{50}	H_{99}	$H_{99.9}$
0	11	9,311,936	11	2	3	4
0.01	11	9,311,936	11	2	3	4
0.02	12	2,851,328	10	2	3	3
0.03	13	2,851,328	10	2	3	3
0.04	13	2,851,328	10	2	3	3
0.05	14	697,792	5	2	2	3
0.075	15	697,792	5	2	2	3
0.10	17	131,072	4	2	2	3
0.15	22	64	2	2	2	2
0.20	27	0	NA	NA	NA	NA
0.25	34	0	NA	NA	NA	NA

The monotonicity of $H_{\max,\delta}$ in δ at fixed N follows simply from the nesting

$$\mathrm{RHF}_{N,\delta_2} \subseteq \mathrm{RHF}_{N,\delta_1} \qquad (\delta_2 > \delta_1).$$

11.1 The thirteen states surviving through lag 9

Exactly thirteen RHC states satisfy

$$d_h \le 0 \qquad (1 \le h \le 9).$$

Exactly two also satisfy $d_{10} \le 0$, and both have $h_* = 11$:

rank	mask	S	h_*	D^-	J	d_{10}	d_{11}
1	247462518	11	11	18	8	-2	8
2	249555574	11	11	20	14	-4	14
3	268299894	13	10	8	4	4	6
4	83952078	-11	10	10	6	6	0
5	201191030	11	10	14	2	2	6
6	251394422	11	10	20	2	2	4
7	251522678	11	10	14	2	2	6
8	251525750	11	10	22	4	4	8
9	262138422	11	10	6	2	2	14
10	264105590	11	10	16	2	2	4
11	267892190	11	10	18	2	2	0
12	268154334	11	10	18	2	2	0
13	268303094	11	10	18	6	6	4

For $N = 110$,

$$M(110) = -5.$$

Both hardest states have $S = 11$, so

$$\sum_{h=1}^{109} d_h = \frac{11^2 - (-5)^2}{2} = 48.$$

Their canonical masks are

$$\boxed{247462518, \qquad 249555574.}$$

They differ only at

$$\boxed{\{43, 83, 86, 89\}},$$

and the four flipped signs have zero total mass, so the endpoint is unchanged.

For the first record,

$$(d_1, \ldots, d_{11}) = (-6, 0, 0, 0, -2, -6, 0, -2, 0, -2, 8),$$

while for the second,

$$(d_1, \ldots, d_{11}) = (-2, 0, -2, -6, -2, 0, -2, -2, 0, -4, 14).$$

Thus the transition

$$13 \longrightarrow 2 \longrightarrow 0$$

is exact: eleven states first become positive at lag 10, whereas the remaining two still have

$$d_{10} = -2, -4$$

and then jump to

$$d_{11} = 8, 14.$$

The pre-witness deficits of the two records are 18 and 20. They are not maximal among the thirteen survivors: one $h_* = 10$ state has $D^- = 22$. Hence a larger accumulated deficit does not by itself imply a later first witness.

11.2 Exact $h = 10$ and $h = 11$ parity systems

At $h = 10$ there are 33 displacement edges and 33 distinct parity forms q. Their Hamming-weight distribution is

$\lvert q\rvert$	1	2	3	4
number	2	13	17	1
edge sign	+	−	+	−

because $h = 10$ is even.

At $h = 11$ there are 32 edges and 31 distinct parity forms. The Hamming-weight distribution, counted with edge multiplicity, is

$\lvert q\rvert$	0	1	2	3	4
number	1	2	11	16	2
edge sign	+	−	+	−	+

because $h = 11$ is odd. The $q = 0$ edge never crosses an admissible multiplicative cut.

The first record has

$$A_{10} - B_{10} = -1, \qquad A_{11} - B_{11} = 4,$$

and the second has

$$A_{10} - B_{10} = -2, \qquad A_{11} - B_{11} = 7,$$

which gives exactly

$$d_{10} = -2, -4, \qquad d_{11} = 8, 14.$$

11.3 Linear rank is already saturated at lag 2

For $N = 110$,

$$\operatorname{rank}_{\mathbb{F}_2} Q_1 = 26, \qquad \operatorname{rank}_{\mathbb{F}_2} Q_2 = 27,$$

but

$$\boxed{\operatorname{rank}_{\mathbb{F}_2}(Q_1 \cup Q_2) = 28,}$$

the full number of free prime-sign bits. Consequently,

$$\operatorname{rank}_{\mathbb{F}_2}(Q_1 \cup \cdots \cup Q_H) = 28 \qquad (H \geq 2).$$

Thus the later lags do not reveal new linear directions in the prime-sign cube. They impose new weighted parity thresholds on an already full-rank family.

For every displacement edge at $N = 110$,

$$\boxed{|q_e| \leq 4.}$$

The parity tests are therefore of degree at most four in the free prime-sign bits. Collisions $q_e = q_{e'}$ are rare: the largest collision count occurs at $h = 63$, where 21 edges give 18 distinct parity forms.

11.4 Complete signed-cut profile of the thirteen survivors

Among the thirteen states, the lags at which all thirteen cuts are strictly positive are

$$\boxed{15, 16, 27, 28, 45, 46, 55, 83, 95,}$$

and the lags at which all thirteen are strictly negative are

$$\boxed{30, 60, 73, 86, 92, 94, 98, 105.}$$

The lags at which no one of the thirteen has a positive cut are

$$\boxed{1, 2, 3, 4, 5, 6, 7, 8, 9, 29, 30, 36, 37, 59, 60, 73, 76, 86, 92, 94, 98, 99, 105, 107.}$$

Some signs are forced already by the displacement graph, independently of multiplicative admissibility: at

$$h = 98, 105, 107$$

every existing edge has negative weight, while at $h = 109$ the unique edge has positive weight.

The complete aggregate data for all 109 lags are:

h	$d_h < 0$	$d_h = 0$	$d_h > 0$	$\min d_h$	$\max d_h$	$\sum d_h$
1	8	5	0	-6	0	-30
2	6	7	0	-4	0	-14
3	7	6	0	-2	0	-14
4	6	7	0	-6	0	-22
5	9	4	0	-6	0	-26
6	8	5	0	-6	0	-28
7	5	8	0	-4	0	-12
8	8	5	0	-6	0	-22
9	5	8	0	-10	0	-28
10	2	0	11	-4	6	28
11	0	3	10	0	14	74
12	3	2	8	-4	6	30
13	3	5	5	-4	6	6
14	1	0	12	-2	8	42
15	0	0	13	18	26	288
16	0	0	13	8	18	156
17	3	4	6	-14	8	0
18	4	3	6	-4	8	14
19	11	0	2	-6	2	-42
20	11	0	2	-10	8	-50
21	2	5	6	-2	16	42
22	1	1	11	-2	8	36
23	1	2	10	-6	8	46
24	6	3	4	-12	4	-20
25	7	5	1	-10	2	-22
26	2	0	11	-6	10	52
27	0	0	13	4	12	102
28	0	0	13	2	18	152
29	11	2	0	-10	0	-72
30	13	0	0	-18	-6	-170
31	9	1	3	-8	6	-26
32	2	0	11	-2	20	82
33	1	1	11	-2	16	56
34	4	1	8	-6	4	8
35	9	1	3	-12	6	-56
36	11	2	0	-22	0	-120
37	11	2	0	-8	0	-48
38	1	2	10	-4	8	38
39	4	1	8	-6	8	30
40	8	2	3	-6	8	-22
41	4	1	8	-10	8	18
42	10	2	1	-14	4	-62
43	1	3	9	-2	6	26
44	0	4	9	0	10	36
45	0	0	13	4	18	152
46	0	0	13	4	10	90
47	11	1	1	-10	2	-76
48	9	1	3	-14	4	-64
49	2	4	7	-8	4	6
50	5	6	2	-6	6	-6
51	3	2	8	-10	8	28
52	1	1	11	-2	10	68
53	4	3	6	-4	6	18
54	3	0	10	-2	10	32
55	0	0	13	2	6	56
56	2	0	11	-2	10	62
57	2	4	7	-4	8	26
58	8	3	2	-6	4	-22
59	12	1	0	-14	0	-110
60	13	0	0	-16	-4	-108
61	8	3	2	-6	6	-22
62	4	4	5	-8	6	-2
63	1	1	11	-6	6	34
64	4	2	7	-2	6	20
65	7	5	1	-8	4	-22
66	1	4	8	-2	6	26
67	2	0	11	-4	10	68
68	5	3	5	-12	2	-20

h	$d_h < 0$	$d_h = 0$	$d_h > 0$	$\min d_h$	$\max d_h$	$\sum d_h$
69	0	3	10	0	10	46
70	6	4	3	-6	4	-12
71	10	1	2	-8	4	-48
72	2	4	7	-2	8	22
73	13	0	0	-8	-2	-60
74	0	2	11	0	8	38
75	0	4	9	0	10	38
76	12	1	0	-12	0	-92
77	0	3	10	0	8	36
78	2	2	9	-6	6	24
79	4	0	9	-6	8	32
80	3	6	4	-8	4	-8
81	12	0	1	-8	2	-74
82	0	1	12	0	8	66
83	0	0	13	2	8	74
84	6	2	5	-6	6	4
85	7	5	1	-4	2	-18
86	13	0	0	-6	-2	-52
87	7	5	1	-4	8	-10
88	2	5	6	-4	12	28
89	1	11	1	-2	8	6
90	1	1	11	-4	6	42
91	1	2	10	-4	8	62
92	13	0	0	-4	-2	-42
93	8	4	1	-2	2	-14
94	13	0	0	-4	-2	-44
95	0	0	13	2	8	46
96	1	5	7	-2	4	14
97	3	0	10	-2	4	16
98	13	0	0	-8	-4	-66
99	12	1	0	-6	0	-54
100	10	2	1	-8	2	-70
101	0	1	12	0	4	26
102	1	1	11	-2	2	20
103	1	2	10	-2	4	38
104	3	0	10	-4	4	14
105	13	0	0	-4	-2	-50
106	1	11	1	-2	2	0
107	12	1	0	-4	0	-44
108	0	11	2	0	2	4
109	0	3	10	0	2	20

12 Complete parity-form statistics for all lags at $N = 110$

For each h, the following table records the number of squarefree displacement edges, the number of distinct parity forms q_e, the number of collisions, the rank of the forms at that lag, the cumulative rank through that lag, and the maximum Hamming weight of a form.

h	$\lvert E_h \rvert$	distinct q	collisions	$\operatorname{rank} Q_h$	cumulative rank	$\max \lvert q \rvert$
1	38	38	0	26	26	4
2	37	37	0	27	28	4
3	35	35	0	25	28	3
4	54	54	0	26	28	4
5	36	35	1	24	28	3
6	35	35	0	27	28	4
7	34	33	1	23	28	4
8	50	50	0	26	28	4
9	38	36	2	23	28	3
10	33	33	0	26	28	4
11	32	31	1	20	28	4

h	$\lvert E_h\rvert$	distinct q	collisions	$\operatorname{rank} Q_h$	cumulative rank	$\max \lvert q\rvert$
12	49	49	0	27	28	4
13	32	31	1	23	28	4
14	33	32	1	26	28	4
15	33	32	1	23	28	3
16	47	46	1	25	28	4
17	30	30	0	20	28	4
18	35	35	0	25	28	4
19	31	31	0	21	28	4
20	45	44	1	25	28	4
21	30	29	1	22	28	3
22	29	29	0	23	28	4
23	32	32	0	22	28	4
24	44	44	0	25	28	3
25	29	29	0	20	28	4
26	28	28	0	23	28	4
27	33	33	0	24	28	3
28	41	39	2	24	28	4
29	27	27	0	20	28	4
30	28	28	0	25	28	3
31	29	29	0	21	28	4
32	41	39	2	25	28	4
33	25	24	1	21	28	3
34	25	25	0	23	28	4
35	26	25	1	20	28	3
36	43	43	0	25	28	4
37	23	23	0	19	28	4
38	25	25	0	24	28	4
39	24	24	0	20	28	3
40	36	35	1	23	28	4
41	23	23	0	19	28	4
42	22	22	0	22	28	3
43	22	22	0	20	28	4
44	32	32	0	23	28	4
45	23	22	1	20	28	3
46	21	21	0	20	28	4
47	21	21	0	17	28	4
48	32	31	1	23	28	3
49	19	19	0	17	28	3
50	23	23	0	22	28	3
51	22	22	0	18	28	3
52	29	29	0	23	28	4
53	17	17	0	15	28	4
54	22	22	0	21	28	3
55	21	21	0	16	28	3
56	28	28	0	23	28	4
57	17	17	0	15	28	3
58	17	17	0	17	28	4
59	20	20	0	19	28	4
60	25	25	0	22	28	3
61	15	15	0	15	28	4
62	16	16	0	16	28	4
63	21	18	3	16	28	3
64	25	25	0	21	28	4
65	16	15	1	15	28	3
66	17	17	0	17	28	3
67	17	17	0	17	28	4
68	23	23	0	21	28	4
69	14	14	0	13	28	3
70	14	14	0	14	28	3
71	16	16	0	15	28	4
72	23	22	1	20	28	3
73	13	13	0	13	28	3
74	13	13	0	13	28	4
75	13	12	1	12	28	2
76	18	18	0	16	28	4
77	11	11	0	11	28	3
78	11	11	0	11	28	3
79	11	11	0	11	28	4

h	$\lvert E_h\rvert$	distinct q	collisions	rank Q_h	cumulative rank	max $\lvert q\rvert$
80	16	16	0	15	28	3
81	11	11	0	11	28	3
82	10	10	0	10	28	4
83	10	10	0	10	28	4
84	14	14	0	14	28	3
85	7	7	0	7	28	3
86	10	10	0	10	28	4
87	9	9	0	9	28	3
88	12	12	0	11	28	4
89	7	7	0	7	28	4
90	9	9	0	9	28	2
91	8	8	0	8	28	3
92	10	10	0	10	28	4
93	6	6	0	6	28	3
94	6	6	0	6	28	4
95	7	7	0	7	28	3
96	8	8	0	8	28	3
97	4	4	0	4	28	3
98	4	4	0	4	28	2
99	6	6	0	6	28	2
100	7	7	0	7	28	2
101	4	4	0	4	28	2
102	4	4	0	4	28	2
103	4	4	0	4	28	3
104	5	5	0	5	28	3
105	3	3	0	3	28	1
106	2	2	0	2	28	2
107	2	2	0	2	28	3
108	2	2	0	2	28	2
109	1	1	0	1	28	2

13 Complete profiles of the two hardest states

13.1 Mask 247462518

h	A_h	B_h	d_h	$C_h(x)$	$C_h(\mu)$
1	8	11	-6	-2	4
2	8	8	0	-3	-3
3	10	10	0	-13	-13
4	10	10	0	-6	-6
5	8	9	-2	4	6
6	6	9	-6	5	11
7	10	10	0	0	0
8	13	14	-2	-4	-2
9	10	10	0	-12	-12
10	6	7	-2	-7	-5
11	9	5	8	12	4
12	14	11	6	19	13
13	9	8	2	6	4
14	9	8	2	-1	-3
15	14	3	22	-3	-25
16	14	8	12	1	-11
17	5	7	-4	-2	2
18	11	11	0	7	7
19	7	9	-4	-1	3
20	9	11	-4	-5	-1
21	4	4	0	-14	-14
22	8	7	2	-1	-3
23	8	6	4	4	0
24	10	10	0	10	10
25	6	6	0	1	1

h	A_h	B_h	d_h	$C_h(x)$	$C_h(\mu)$
26	7	2	10	10	0
27	10	6	8	-3	-11
28	15	9	12	5	-7
29	3	6	-6	3	9
30	3	10	-14	2	16
31	4	6	-4	1	5
32	11	10	2	-5	-7
33	6	5	2	-9	-11
34	7	5	4	5	1
35	4	5	-2	-4	-2
36	8	12	-8	7	15
37	5	5	0	5	5
38	8	6	4	3	-1
39	7	5	4	2	-2
40	8	10	-4	-4	0
41	5	4	2	5	3
42	4	6	-4	4	8
43	4	3	2	4	2
44	10	9	2	-2	-4
45	8	2	12	-1	-13
46	6	4	4	-3	-7
47	1	5	-8	-7	1
48	7	10	-6	2	8
49	6	5	2	3	1
50	6	6	0	-1	-1
51	7	6	2	-2	-4
52	7	2	10	5	-5
53	5	6	-2	-5	-3
54	5	4	2	6	4
55	6	4	4	3	-1
56	7	6	2	4	2
57	6	4	4	-1	-5
58	2	5	-6	-9	-3
59	2	8	-12	-6	6
60	5	8	-6	5	11
61	5	5	0	3	3
62	3	2	2	2	0
63	4	2	4	-3	-7
64	5	5	0	-3	-3
65	3	4	-2	-2	0
66	4	2	4	7	3
67	5	1	8	9	1
68	4	3	2	5	3
69	3	2	2	-4	-6
70	4	6	-4	-4	0
71	3	6	-6	-4	2
72	5	5	0	3	3
73	1	4	-6	-1	5
74	6	4	4	1	-3
75	6	3	6	-1	-7
76	2	6	-8	-6	2
77	4	3	2	-1	-3
78	4	1	6	9	3
79	3	1	4	5	1
80	5	5	0	2	2
81	1	5	-8	-7	1
82	4	2	4	0	-4
83	3	0	6	0	-6
84	3	4	-2	0	2
85	1	2	-2	1	3
86	2	3	-2	0	2
87	2	2	0	-3	-3
88	2	3	-2	-6	-4
89	0	0	0	-3	-3
90	3	0	6	9	3
91	5	1	8	6	-2
92	1	3	-4	-2	2
93	0	1	-2	-4	-2

h	A_h	B_h	d_h	$C_h(x)$	$C_h(\mu)$
94	0	2	-4	0	4
95	2	1	2	-1	-3
96	2	2	0	2	2
97	1	0	2	2	0
98	0	2	-4	0	4
99	0	2	-4	-4	0
100	0	4	-8	-5	3
101	2	1	2	0	-2
102	1	0	2	4	2
103	2	0	4	4	0
104	2	1	2	1	-1
105	0	2	-4	-1	3
106	1	1	0	0	0
107	0	2	-4	-2	2
108	1	1	0	0	0
109	1	0	2	1	-1

13.2 Mask 249555574

h	A_h	B_h	d_h	$C_h(x)$	$C_h(\mu)$
1	9	10	-2	2	4
2	9	9	0	-3	-3
3	9	10	-2	-15	-13
4	10	13	-6	-12	-6
5	8	9	-2	4	6
6	8	8	0	11	11
7	10	11	-2	-2	0
8	13	14	-2	-4	-2
9	10	10	0	-12	-12
10	6	8	-4	-9	-5
11	11	4	14	18	4
12	13	13	0	13	13
13	10	7	6	10	4
14	9	7	4	1	-3
15	14	3	22	-3	-25
16	14	10	8	-3	-11
17	6	5	2	4	2
18	10	9	2	9	7
19	8	11	-6	-3	3
20	9	11	-4	-5	-1
21	5	2	6	-8	-14
22	8	6	4	1	-3
23	10	6	8	8	0
24	12	15	-6	4	10
25	6	6	0	1	1
26	7	4	6	6	0
27	10	6	8	-3	-11
28	17	8	18	11	-7
29	4	7	-6	3	9
30	3	12	-18	-2	16
31	4	7	-6	-1	5
32	11	9	4	-3	-7
33	8	5	6	-5	-11
34	7	5	4	5	1
35	4	7	-6	-8	-2
36	8	14	-12	3	15
37	3	5	-4	1	5
38	7	7	0	-1	-1
39	9	5	8	6	-2
40	8	8	0	0	0
41	5	4	2	5	3
42	2	6	-8	0	8

h	A_h	B_h	d_h	$C_h(x)$	$C_h(\mu)$
43	5	3	4	6	2
44	11	9	4	0	-4
45	10	2	16	3	-13
46	6	3	6	-1	-7
47	1	5	-8	-7	1
48	5	10	-10	-2	8
49	6	5	2	3	1
50	5	6	-2	-3	-1
51	7	5	4	0	-4
52	6	5	2	-3	-5
53	5	6	-2	-5	-3
54	6	4	4	8	4
55	6	4	4	3	-1
56	9	6	6	8	2
57	6	4	4	-1	-5
58	2	3	-2	-5	-3
59	2	8	-12	-6	6
60	5	10	-10	1	11
61	4	5	-2	1	3
62	4	1	6	6	0
63	3	2	2	-5	-7
64	5	6	-2	-5	-3
65	3	5	-4	-4	0
66	4	5	-2	1	3
67	7	2	10	11	1
68	4	3	2	5	3
69	5	2	6	0	-6
70	4	6	-4	-4	0
71	3	7	-8	-6	2
72	5	4	2	5	3
73	3	4	-2	3	5
74	5	4	2	-1	-3
75	4	3	2	-5	-7
76	2	7	-10	-8	2
77	3	3	0	-3	-3
78	4	3	2	5	3
79	3	1	4	5	1
80	5	3	4	6	2
81	2	4	-4	-3	1
82	3	1	4	0	-4
83	3	0	6	0	-6
84	2	3	-2	0	2
85	1	1	0	3	3
86	2	4	-4	-2	2
87	2	3	-2	-5	-3
88	3	3	0	-4	-4
89	0	0	0	-3	-3
90	3	0	6	9	3
91	5	1	8	6	-2
92	1	3	-4	-2	2
93	0	1	-2	-4	-2
94	0	2	-4	0	4
95	2	1	2	-1	-3
96	2	2	0	2	2
97	1	0	2	2	0
98	0	2	-4	0	4
99	0	2	-4	-4	0
100	0	4	-8	-5	3
101	2	1	2	0	-2
102	1	0	2	4	2
103	2	0	4	4	0
104	2	1	2	1	-1
105	0	2	-4	-1	3
106	1	1	0	0	0
107	0	2	-4	-2	2
108	1	1	0	0	0
109	1	0	2	1	-1

14 Bibliographic assessment of the signed-cut formulation

The identity

$$d_h/2 = \mathrm{Cut}_h(D)$$

is elementary once the discrepancy set is introduced, and signed cuts themselves are classical [6, 7]. Likewise, finite aperiodic autocorrelation is classical [5]. The multiplicative-function literature studies correlations and sign patterns by analytic and probabilistic methods [4, 3, 2].

The combination used here is more specific:

$$\mathcal{D}_H(N) = \{D = Ab : \mathrm{Cut}_h(D) \le 0,\ 1 \le h \le H\},$$

possibly intersected with an endpoint class such as RHC. Equivalently, on the free prime-sign cube,

$$\mathcal{B}_H(N) = \left\{ b : \sum_{q \cdot b = 1} m_h(q)(-1)^{|q|+h+1} \le 0,\ 1 \le h \le H \right\}.$$

The nested filtration

$$\mathcal{B}_1 \supseteq \mathcal{B}_2 \supseteq \cdots$$

and the first positive threshold

$$h_*(b) = \min \left\{ h : \sum_{q \cdot b = 1} m_h(q)(-1)^{|q|+h+1} > 0 \right\}$$

are the combinatorial objects measured by the exhaustive computation.

No exact antecedent for this simultaneous Möbius-signed displacement-cut filtration, restricted to the multiplicative squarefree completion code, was located in the literature searches performed for this work. The statement is deliberately narrower than a claim that signed cuts, autocorrelation, parity functions, or multiplicative correlations are new.

15 Conclusions: original, proved, and useful content

Within the bibliography surveyed here, the original contribution supported by complete proofs and exact computation is the following finite combinatorial package.

First, the short-correlation comparison against the true Möbius word is represented exactly as a simultaneous signed-cut problem:

$$\boxed{\frac{C_h(x) - C_h(\mu)}{2} = \mathrm{Cut}_{G_h}(D(x)).}$$

The admissible discrepancy sets are not arbitrary cuts; they form the binary code image

$$\boxed{\mathbf{1}_D = Ab \pmod 2}$$

generated by odd-prime divisibility. Every displacement edge is activated by a parity equation

$$q_e \cdot b = 1,$$

and its sign is determined internally by

$$\boxed{-\mu(n)\mu(n+h) = (-1)^{|q_e|+h+1}.}$$

Consequently the first witness is an explicit threshold of weighted parity tests,

$$\boxed{h_*(b) = \min\left\{ h : \sum_{q\cdot b=1} m_h(q)(-1)^{|q|+h+1} > 0 \right\}.}$$

No exact antecedent for this simultaneous displacement-cut filtration under multiplicative admissibility was found in the targeted literature search.

Second, the exhaustive $N = 110$ instance is completely resolved for the tested family:

$$2^{28} = 268{,}435{,}456$$

states,

$$9{,}311{,}936$$

RHC states, and the exact final survivor chain

$$\boxed{13 \to 2 \to 0}$$

at lags $9, 10, 11$, with

$$\boxed{H_{\max,0}(110) = 11.}$$

The two final states, their complete 109-lag cut profiles, and the complete parity-form statistics for every lag are given in the tables above. These data are not used to assert an asymptotic law; their value is that they expose the finite object to be explained combinatorially.

The qualified novelty claim of this paper is therefore the simultaneous signed displacement-cut formulation restricted to the multiplicative Möbius completion code, together with its exact parity-threshold reduction and complete finite $N = 110$ resolution.

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